

Johan Ajnabi

PhD Candidate | Wound Healing, Mechanobiology and Epigenetics

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem)

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Summary

Mechanobiology and epigenetics researcher studying how mechanical forces regulate chromatin organization and cell fate transitions during tissue repair. My work identifies cytoskeleton-driven control of DNMT3a localization as a mechanism linking tissue mechanics to epigenetic reprogramming. I combine cell biology, signaling, and molecular biology tools to investigate how tissue architecture shapes regeneration and disease. I am interested in advancing research at the interface of mechanobiology, epithelial biology, and tissue-scale control of cell states.

Education

PhD Candidate, BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem) 2019 - Present
Advisor: Prof. Colin Jamora

PhD thesis: Regulation of DNMT3A localization in the cutaneous wound healing response

M.Sc. in Molecular Biology and Biotechnology, ICAR-Indian Agricultural Research Institute 2019

B.Sc. (Hons.) in Agriculture, Bidhan Chandra Krishi Viswavidyalaya 2017

Research Experience

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem), India 2019-Present
Research Scholar (PhD)

Advisor: Prof. Colin Jamora

Mechanotransduction and epigenetic regulation (Primary project)

- Identified a mechanosensitive regulation mechanism of DNMT3a nucleocytoplasmic shuttling, establishing a link between cytoskeletal dynamics and epigenetic reprogramming during wound repair
- Integrated live-cell imaging, biochemical assays, and genetic perturbations to define ERK-dependent mechanotransduction pathways
- Led first-author study (bioRxiv: [Ajnabi et al., 2026](#)).

Stem cell signaling and niche regulation

- Contributed to the discovery of a Mindin-integrin-STAT3 signaling axis that sustains keratinocyte stemness through integrin endocytosis. Co-author preprint (bioRxiv: [Dam et al., 2025](#)).
- Contributed to work showing how autocrine Mindin signaling maintains stem and progenitor states in skin epithelium and carcinoma cells. (Published in Cell Reports: [Badarinath et al., 2022](#)).

Tissue-scale and translational biology (vitiligo, fibrosis)

- Performed spatial transcriptomic profiling of vitiligo skin before and after NB-UVB treatment and validated key findings by immunofluorescence and qPCR (Published in Clinical & Translational Immunology: [Dutta et al., 2026](#)).
- Studied extracellular matrix regulation of cutaneous fibrosis using transgenic mouse models. (Published in Journal of Investigative Dermatology: [Rana et al., 2023](#)).

Host-pathogen interactions

- Investigated LL-37-mediated inhibition of SARS-CoV-2 entry using mammalian cell systems, pseudovirus assays, and flow cytometry (Published, Frontiers in Immunology: [Bhatt et al., 2023](#)).

ICAR-National Institute for Plant Biotechnology, India

2017-2019

M.Sc. dissertation

Advisor: Dr. Monika Dalal

- Characterized promoter cis-regulatory regions controlling *PM19* expression in wheat using cloning, transient GUS assays, plant tissue culture, transformation, and sequence analysis (Thesis: [Ajnabi J, 2019](#))

Publications

First-Author & Lead Contributions

- **Ajnabi J**, Dam B, Gupta E, Saha T, et al. *Actin-dependent mechanotransduction controls nucleocytoplasmic partitioning of DNMT3A through ERK1/2 signaling during cutaneous wound healing.*
bioRxiv, 2026 (preprint)

Collaborative Publications

- Dutta A, Gupta D, **Ajnabi J**, et al. *Spatial transcriptomic analysis of the immune landscape following NB-UVB treatment of vitiligo skin.*
Clinical & Translational Immunology, 2026
- Dutta S, Islam Z, Das S, Barman A, Chowdhury M, Mondal BP, **Ajnabi J**, et al. *Harmonizing plant resilience: membrane lipid dynamics in response to abiotic stresses.*
Discover Plants, 2025
- Bhatt T, Dam B, Khedkar SU, Lall S, Pandey S, Kataria S, **Ajnabi J**, et al. *Niacinamide enhances cathelicidin-mediated SARS-CoV-2 membrane disruption.*
Frontiers in Immunology, 2023
- Rana I, Kataria S, Tan TL, Hajam EY, Kashyap DK, Saha D, **Ajnabi J**, et al. *Mindin (SPON2) is essential for cutaneous fibrogenesis in a mouse model of systemic sclerosis.*
Journal of Investigative Dermatology, 2023
- Badarinath K, Dam B, Kataria S, Zirmire RK, Dey R, Kansagara G, **Ajnabi J**, et al. *Snail maintains the stem/progenitor state of skin epithelial cells and carcinomas through autocrine Mindin signaling.*
Cell Reports, 2022

Preprints (Co-authored)

- Dam B, **Ajnabi J**, et al. *Mindin-mediated α M-integrin endocytosis activates STAT3 to maintain keratinocyte stemness.*
bioRxiv, 2025

Fellowships and Awards

- **Paeonia Foundation Travel Award** – Selected to present research at *Mechanobiology in Health and Disease Conference*, National University of Singapore, 2023
- **ASRB-NET (Agricultural Biotechnology)** – Qualified national eligibility test for lectureship, 2020
- **CSIR-NET Junior Research Fellowship (JRF) (AIR 20)** – National-level fellowship awarded to top-ranked candidates by the Council of Scientific and Industrial Research (CSIR), India, 2019
- **ICMR Junior Research Fellowship (JRF)** – Competitive national fellowship in Life Sciences by the Indian Council of Medical Research (ICMR), 2019
- **GATE XL (AIR 8)** – All India Rank 8 in Graduate Aptitude Test in Engineering (Life Sciences), 2019
- **DBT-BCIL Fellowship** – National-level fellowship awarded by the Department of Biotechnology, Government of India, 2019
- **ICAR AICE-JRF/SRF (AIR 3)** – All India Rank 3 in national examination for agricultural biotechnology research fellowships by the Indian Council of Agricultural Research (ICAR), 2019
- **ICAR AIEEA-PG (AIR 1)** – All India Rank 1 in national entrance examination for postgraduate studies in agricultural biotechnology, 2017

Selected Presentations

- **Poster** – “Epigenetic and mechanical regulation of the cutaneous wound healing response” at *Mechanobiology in Health and Disease Conference*, National University of Singapore, Singapore, 2023
- **Poster** – “Role of DNMT3A in the cutaneous wound healing response using a mouse model” at *10th International Conference of Laboratory Animal Scientists’ Association (LASA)*, Hyderabad, India, 2022
- **Poster** – “Identification and characterization of stress-responsive PM19 promoter from wheat” at *National Agricultural Science Congress*, New Delhi, India, 2019

Skills

Experimental Expertise

- Experimental design and hypothesis-driven research
- Experimental systems: mouse models and colony management, primary mammalian cell culture
- Imaging and tissue analysis: confocal & multiphoton microscopy, Flow cytometry (FACS)
- Molecular and cell biology: genetic perturbation, transfection, qPCR, western blotting, protein purification, flow cytometry, pseudovirus generation, BSL-2 workflows

Computational & Analytical

- R (basic), Python (basic),
- Image analysis (FIJI/ImageJ)

Mentoring & Collaboration

- Mentored junior researchers in primary cell culture, molecular biology techniques, and experimental design, including training in data interpretation and scientific communication (presentations and writing)
- Collaborated on interdisciplinary projects spanning mechanobiology, immunology, and spatial transcriptomics, integrating experimental and clinical datasets
- Contributed to multi-author studies across epithelial biology, fibrosis, and host–pathogen interactions, supporting experimental design, data generation, and analysis

Languages

- Bengali (Native)
- English (Advanced)
- Hindi (Limited working proficiency)

References

Prof. Colin Jamora

Senior Professor

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Pradesh - 201314, India

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Prof. Dasaradhi Palakodeti

Professor

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